AI Application R&D Engineer · BenQ Materials

# Shan-Cheng Chang 張善程

LLM Agent Engineering · AI Data Analytics · Document AI

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## — CORE SKILLS

**Agent Engineering (live)**
Skills (Claude Code style) · Sandboxing · RBAC / Audit Log

**Agent Prototypes (PoC)**
Agent Harness · Function Calling · MCP · Context Management · Knowledge Graph (PostgreSQL + pgvector)

**Evaluation & Validation**
Golden Set · Code Validators

**LLM**
Azure OpenAI · Gemini API · Prompt Engineering · APO

**Document AI / ML / CV**
Azure Document Intelligence · OCR + LLM Extraction · YOLOv8 · PyTorch

**Backend & Web**
Python · FastAPI · Node.js · React · JavaScript · PostgreSQL · SQL Server

**DevOps & Tools**
Docker · GCP · Azure · Git · Claude Code

**Security**
AES-256-GCM · JWT · AD / Entra ID · ISO 27001-aligned

## — EDUCATION

### National Taiwan University
M.S. Biomechatronics Eng. (AIoT / CV) · 2021 – 2023

### Taipei Tech (NTUT)
B.S. Mechanical Engineering · 2017 – 2021

AI application engineer who designs, validates and maintains enterprise AI systems: architect of BenQ Materials' AI data-analysis platform (in daily production use) and its document-AI stack, built from zero. Every system is checked against a test set or the system it replaces before go-live.

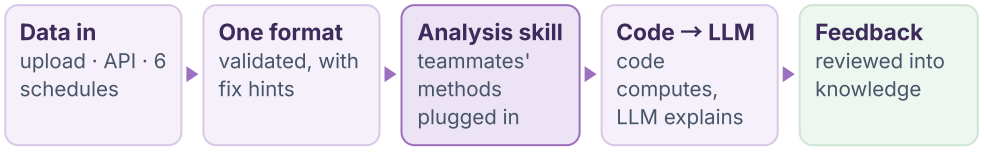
| 4.15M NTD | 6 | ~1M NTD | 1,500+ h |
|---|---|---|---|
| AUTOML AVOIDED / 2 YRS | LINE FLOWS ON PLATFORM | BENEFIT / YR (3 OCR) | SAVED / YR (3 OCR) |

## — EXPERIENCE

### AI Application R&D Engineer
Dec 2024 – Present

BenQ Materials Corp. (明基材料) · Digital Technology Center, Intelligent Application Dev. Sec. · Taoyuan

- **Enterprise Brain → AI data-analysis platform** (system architect & lead developer) — used daily by the BML line, replacing a purchased AutoML package (**4.15M NTD / 2 yrs**); 6 flows cut over with zero downtime. FastAPI + Azure OpenAI + Docker sandbox, on-prem.




Data → analysis → knowledge. Purple box: the shared skill I designed.

| | Before (n8n) | After (one skill) |
|---|---|---|
| Analysis logic | 6 flows, ~550 nodes | **1 skill, 3 entry points** |
| New method | edit every flow | **teammate's logic, wired once** |
| Migration check | — | **Old = new: Excel 9/9 cells; same PDFs** |

  - **Why a skill** — hard-coding would work today; a Claude Code–format skill lets a future chat LLM call the same methods. Analysis already runs in a Docker sandbox that chat will reuse.
  - **Status** — analysis platform live; now building an OCAP text knowledge base (K1, incl. my 6 flows) and a user-feedback base (K2) before chat; MCP and knowledge graph are prototypes. Full architecture on my website.
- **Document-AI applications (OCR + LLM)** — built the company's platform from zero (Azure Document Intelligence + Azure OpenAI); every system scored field-by-field in code on a hand-labelled test set before go-live.

| System | Accuracy (test set) | Impact |
|---|---|---|
| **Quotation OCR** | validity **51% → 100%** (29 docs) | 30 → 10 s; 758+ docs |
| **Invoice OCR (TW/CN)** | **31/32**, **31/31** fully correct | 1,447 h/yr; CN: 3 sites |
| **Health-check OCR** | **97.4%** fields (42 × 46) | AES-256, AD; ISO 27001-aligned |
| **APO-OCR prompt tuning** | validating (32-invoice set) | team tool, pre-launch |

## — LANGUAGES

Mandarin Chinese — Native
English — Professional (TOEIC 790)
German — A2

## — CERTIFICATIONS

Company Digital Talent L3 (Expert)
Military service completed

## — EARLIER PROJECT

**Autonomous navigation with Autoware / ROS**
B.S. capstone, NTUT · 2020 – 2021
Indoor/outdoor self-driving platform (mapping, localization, path planning) with an ML mask-detection module.

## — PUBLICATIONS & PATENT

Chou, C.-Y., **Chang, S.-C.**, et al. (2023). Development of AIoT system for facility asparagus cultivation. *Computers and Electronics in Agriculture*, 206:107665. *(co-first author with advisor)*

**Chang, S.-C.**, et al. (2022). AIoT based pest counting system for asparagus cultivation. *Proc. 10th ISMAB*.

Taiwan Invention Patent TW I842416 B — Crop Harvester, 2024.

## — EXPERIENCE (CONT.)

### BenQ Materials — other systems

Dec 2024 – Present

- **Google Cloud Platform — supplier questionnaire management system** (live) — 52 APIs, built solo in 4.5 months, handed over; React 19 + Cloud Functions + Entra ID.
- **ESG carbon platform refactor & handover** — import moved to a web upload run by the user department; backend split into one module per page (APIs 104 → 42), 55 copy-pasted chart files merged into 5 shared components; every step verified output-identical.

### Shift Supervisor (值班股長)

Jan 2024 – Nov 2024

China Steel Corporation (中國鋼鐵) · Cold-Rolling Mill · Kaohsiung, Taiwan

- **Ran shift operations** on the cold-rolled coil line at **95% yield**.
- **Anomaly analysis** — SPC, trend and correlation analysis for root causes of defects and downtime; daily anomaly reports (Python / Excel).

## — RESEARCH & SELECTED PROJECTS

**AIoT Pest Monitoring & Bee Pollination (M.S. thesis, NTU)** 2021 – 2023

Lead developer of greenhouse AIoT systems: bee tracking *99.5% mAP*, pest counting *95.8%* (YOLOv8, StrongSORT, Mask R-CNN).

**Universal OCR Platform — no-code OCR builder (side project)** 2025 – 2026

Define fields and prompts, deploy as an OCR REST API in one click (FastAPI · React · Cloud Run).